Complete Mathematical Derivation: FDM for Spherical Diffusion-Induced Stresses

Reproducing Cheng & Verbrugge (2009), *J. Power Sources* 190, 453–460

1 Problem Statement

We solve for the concentration profile $C(r,t)$ inside a sphere of radius R undergoing solid-state diffusion, then compute the resulting mechanical stresses.

1.1 Governing PDE — Fick's Second Law in Spherical Coordinates

$$\frac{\partial C}{\partial t} = \frac{D}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial C}{\partial r}\right) \tag{1}$$

where D is the diffusion coefficient (constant), r is the radial coordinate, and $C(r,t)$ is the solute concentration.

Expanding the spatial operator:

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial C}{\partial r}\right) = \frac{\partial^2 C}{\partial r^2} + \frac{2}{r}\frac{\partial C}{\partial r} \tag{2}$$

Proof: Apply the product rule to $\frac{\partial}{\partial r}(r^2\frac{\partial C}{\partial r}) = 2r\frac{\partial C}{\partial r} + r^2\frac{\partial^2 C}{\partial r^2}$, then divide by r^2.

1.2 Dimensionless Variables

Define:

- $x = r/R$ (normalized radius, $0 \le x \le 1$)
- $\tau = Dt/R^2$ (dimensionless time)

Potentiostatic: $c = (C - C_0)/(C_R - C_0)$
Galvanostatic: $c = (C - C_0)/(IR/FD)$
where C_0 is the initial concentration, C_R is the surface concentration (potentiostatic), I is the current density, and F is Faraday's constant.

The PDE becomes:

$$\frac{\partial c}{\partial \tau} = \frac{\partial^2 c}{\partial x^2} + \frac{2}{x}\frac{\partial c}{\partial x} \tag{3}$$

1.3 Boundary Conditions

At the center ($x = 0$): Symmetry requires $\left.\frac{\partial c}{\partial x}\right|_{x=0} = 0$

At the surface ($x = 1$):

- Potentiostatic: $c(1, \tau) = 1$
- Galvanostatic: $\left.\frac{\partial c}{\partial x}\right|_{x=1} = 1$

Initial condition: $c(x, 0) = 0$

1.4 Singularity at $x = 0$

The term $\frac{2}{x}\frac{\partial c}{\partial x}$ is singular at $x = 0$. We resolve this using **L'Hôpital's rule**:

$$\lim_{x\to 0} \frac{2}{x}\frac{\partial c}{\partial x} = 2\lim_{x\to 0} \frac{\partial^2 c}{\partial x^2} = 2\left.\frac{\partial^2 c}{\partial x^2}\right|_{x=0}$$

Therefore, at $x = 0$:

$$\left.\frac{\partial c}{\partial \tau}\right|_{x=0} = 3\left.\frac{\partial^2 c}{\partial x^2}\right|_{x=0} \tag{4}$$

2 Finite Difference Discretization

2.1 Grid Setup

Divide $[0, 1]$ into N equal intervals:

- $x_i = i \cdot \Delta x$ for $i = 0, 1, \ldots, N$
- $\Delta x = 1/N$
- $c_i^n \approx c(x_i, n \cdot \Delta\tau)$

2.2 Standard Finite Difference Approximations

Second derivative (central difference):

$$\left.\frac{\partial^2 c}{\partial x^2}\right|_{x_i} \approx \frac{c_{i+1} - 2c_i + c_{i-1}}{\Delta x^2} \tag{5}$$

First derivative (central difference):

$$\left.\frac{\partial c}{\partial x}\right|_{x_i} \approx \frac{c_{i+1} - c_{i-1}}{2\Delta x} \tag{6}$$

Both are second-order accurate: $\mathcal{O}(\Delta x^2)$.

2.3 Spatial Operator at Interior Points ($i = 1, \ldots, N-1$)

Substituting (5) and (6) into (3):

$$\mathcal{L}(c)_i = \frac{c_{i+1} - 2c_i + c_{i-1}}{\Delta x^2} + \frac{2}{x_i} \cdot \frac{c_{i+1} - c_{i-1}}{2\Delta x}$$

Since $x_i = i \cdot \Delta x$:

$$\frac{2}{x_i \cdot 2\Delta x} = \frac{1}{i \cdot \Delta x^2}$$

Therefore:

$$\mathcal{L}(c)_i = \frac{1}{\Delta x^2} \left[\frac{i-1}{i} c_{i-1} - 2c_i + \frac{i+1}{i} c_{i+1} \right] \tag{7}$$

2.4 Spatial Operator at the Center ($i = 0$)

From Eq. (4): $\left.\frac{\partial c}{\partial \tau}\right|_{x=0} = 3 \left.\frac{\partial^2 c}{\partial x^2}\right|_{x=0}$

Using the symmetry condition $c_{-1} = c_1$ (ghost point):

$$\left.\frac{\partial^2 c}{\partial x^2}\right|_{x=0} \approx \frac{c_1 - 2c_0 + c_{-1}}{\Delta x^2} = \frac{2(c_1 - c_0)}{\Delta x^2} \tag{8}$$

Therefore:

$$\mathcal{L}(c)_0 = \frac{6(c_1 - c_0)}{\Delta x^2} \tag{9}$$

2.5 The Crank-Nicolson Scheme

The Crank-Nicolson method averages the spatial operator at times n and $n+1$:

$$\frac{c_i^{n+1} - c_i^n}{\Delta \tau} = \frac{1}{2} \left[\mathcal{L}(c^{n+1})_i + \mathcal{L}(c^n)_i \right] \tag{10}$$

Properties:

- **Unconditionally stable** (no restriction on $\Delta\tau/\Delta x^2$)
- **Second-order accurate** in both space and time: $\mathcal{O}(\Delta x^2 + \Delta \tau^2)$
- **Implicit** — requires solving a linear system at each time step

Define $s = \Delta\tau/(2\Delta x^2)$.

2.5.1 Interior Points ($i = 1, \ldots, N-1$)

Substituting (7) into (10) and rearranging:

Left-hand side (unknowns at $n+1$):

$$-s\frac{i-1}{i} c_{i-1}^{n+1} + (1+2s)\, c_i^{n+1} - s\frac{i+1}{i} c_{i+1}^{n+1}$$

Right-hand side (knowns at n):

$$s\frac{i-1}{i} c_{i-1}^{n} + (1-2s)\, c_i^{n} + s\frac{i+1}{i} c_{i+1}^{n}$$

Derivation step-by-step: Starting from Eq. (10):

$$c_i^{n+1} - c_i^n = s\left[\frac{i-1}{i}(c_{i-1}^{n+1} + c_{i-1}^n) - 2(c_i^{n+1} + c_i^n) + \frac{i+1}{i}(c_{i+1}^{n+1} + c_{i+1}^n)\right]$$

Move all $n+1$ terms to the left:

$$c_i^{n+1} - s\frac{i-1}{i}c_{i-1}^{n+1} + 2s\,c_i^{n+1} - s\frac{i+1}{i}c_{i+1}^{n+1} = c_i^n + s\frac{i-1}{i}c_{i-1}^n - 2s\,c_i^n + s\frac{i+1}{i}c_{i+1}^n$$

This gives the tridiagonal system $\mathbf{A}\,\mathbf{c}^{n+1} = \mathbf{B}\,\mathbf{c}^n + \mathbf{b}$.

2.5.2 Center Point ($i = 0$)

From Eq. (9):

$$(1+6s)\,c_0^{n+1} - 6s\,c_1^{n+1} = (1-6s)\,c_0^n + 6s\,c_1^n \tag{11}$$

2.5.3 Summary of Matrix Coefficients

Table 1: Matrix A Coefficients

Location	Sub-diagonal	Diagonal	Super-diagonal
$i = 0$	—	$1+6s$	$-6s$
$i = 1 \ldots N-1$	$-s(i-1)/i$	$1+2s$	$-s(i+1)/i$

The **B matrix** (RHS) has the same structure with opposite signs on s:

Table 2: Matrix B Coefficients

Location	Sub-diagonal	Diagonal	Super-diagonal
$i = 0$	—	$1-6s$	$6s$
$i = 1 \ldots N-1$	$s(i-1)/i$	$1-2s$	$s(i+1)/i$

3 Boundary Condition Implementation

3.1 Potentiostatic ($c(1,\tau) = 1$)

Simply set:

- $A_{N,N} = 1$, all other entries in row N are zero
- RHS: $b_N = 1$

3.2 Galvanostatic ($\partial c/\partial x|_{x=1} = 1$)

Ghost-point approach: Introduce a fictitious point c_{N+1} beyond the boundary:

$$\left.\frac{\partial c}{\partial x}\right|_{x=1} = \frac{c_{N+1} - c_{N-1}}{2\Delta x} = 1 \quad \Rightarrow \quad c_{N+1} = c_{N-1} + 2\Delta x \tag{12}$$

Apply the **interior stencil at** $i = N$ and substitute c_{N+1}. Let $a_{\text{sub}} = -s(N-1)/N$ and $a_{\text{sup}} = -s(N+1)/N$.

After substitution:

- Coefficient of c_{N-1}: $a_{\text{sub}} + a_{\text{sup}} = -2s$
- Diagonal: $1 + 2s$ (unchanged)
- Constant: $-a_{\text{sup}} \cdot 2\Delta x$ moves to the RHS

For the B matrix similarly with $b_{\text{sub}} = s(N-1)/N$, $b_{\text{sup}} = s(N+1)/N$:

$$\text{RHS constant} = (-a_{\text{sup}} + b_{\text{sup}}) \cdot 2\Delta x = \frac{2s(N+1)}{N} \cdot 2\Delta x = \frac{2(N+1)\Delta\tau}{N\Delta x}$$

4 Stress Calculations

4.1 Volume-Averaged Concentration

$$\bar{c}(x) = \frac{3}{x^3}\int_0^x x'^2\, c(x')\, dx' \tag{13}$$

At $x = 0$: $\bar{c}(0) = c(0)$ (by L'Hôpital). Evaluated numerically via the trapezoidal rule.

4.2 Dimensionless Stresses (from Eq. 3 of paper)

Normalized by $E\Omega\Delta c/[3(1-\nu)]$:

$$\tilde{\sigma}_r = \frac{2}{3}[\bar{c}(1) - \bar{c}(x)], \quad \tilde{\sigma}_\theta = \frac{1}{3}[2\bar{c}(1) + \bar{c}(x) - 3c(x)] \tag{14}$$

$$\tilde{\sigma}_{\text{shear}} = \frac{\tilde{\sigma}_r - \tilde{\sigma}_\theta}{2} \tag{15}$$

Derivation of (15): From paper Eq. (3): $\sigma_r = \frac{2E\Omega}{9(1-\nu)}[C_{\text{av}}(R) - C_{\text{av}}(r)]$. Dividing by $\frac{E\Omega\Delta c}{3(1-\nu)}$ gives:

$$\tilde{\sigma}_r = \frac{2}{9} \cdot 3 \cdot \frac{C_{\text{av}}(R) - C_{\text{av}}(r)}{\Delta c} = \frac{2}{3}[\bar{c}(1) - \bar{c}(x)]$$

5 Strain Energy

5.1 Strain Energy Density (Eq. 6 of paper)

$$e(r) = \frac{\sigma_r^2 + 2\sigma_\theta^2 - 2\nu\sigma_\theta(2\sigma_r + \sigma_\theta)}{2E} \tag{16}$$

5.2 Dimensionless Total Strain Energy (Eqs. 19, 28)

$$\tilde{E}_T = \frac{E_T}{2\pi R^3 E[\Omega\Delta c/(3(1-\nu))]^2} = \int_0^1 \left[\tilde{\sigma}_r^2 + 2\tilde{\sigma}_\theta^2 - 2\nu\tilde{\sigma}_\theta(2\tilde{\sigma}_r + \tilde{\sigma}_\theta)\right] x^2\, dx \tag{17}$$

6 Analytical Solutions for Validation

6.1 Potentiostatic (Eq. 11 — Carslaw & Jaeger [32])

$$c(x,\tau) = 1 + 2\sum_{n=1}^{\infty} \frac{(-1)^n}{n\pi x} \sin(n\pi x)\, e^{-n^2\pi^2\tau} \tag{18}$$

6.2 Galvanostatic (Eq. 20)

$$c(x,\tau) = 3\tau + \frac{x^2}{2} - \frac{3}{10} - \frac{2}{x}\sum_{n=1}^{\infty} \frac{\sin(\alpha_n x)}{\alpha_n^2 \sin(\alpha_n)} e^{-\alpha_n^2\tau} \tag{19}$$

where α_n are positive roots of $\tan(\alpha) = \alpha$.

7 Implementation Summary

Aspect	**Choice**	**Rationale**
Time scheme	Crank-Nicolson	Unconditionally stable, 2nd-order
Spatial scheme	Central differences	2nd-order, symmetric
Center singularity	L'Hôpital + ghost point	Avoids division by zero
Neumann BC	Ghost point into interior	Maintains 2nd-order accuracy
Linear solve	Dense `numpy.linalg.solve`	Simple; tridiagonal solver possible
Integration	Trapezoidal rule	Sufficient for smooth profiles
Grid	$N = 200$	Errors $\sim \mathcal{O}(10^{-5})$ vs analytical

References

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